

Decomposing Lexicon-Assisted Correction for Isolated Handwritten Word Recognition on IAM: Effects of Lexicon Coverage and Frequency-Based Ranking

Nur Banu Oğur*

*Department of Software Engineering, Faculty of Computer and Information Sciences,
Sakarya University, 54050 Sakarya, Türkiye
nbogur@sakarya.edu.tr*

Rıdvan Dursun

*Department of Software Engineering, Faculty of Computer and Information Sciences,
Sakarya University, 54050 Sakarya, Türkiye
ridvan.dursun@ogr.sakarya.edu.tr*

Berhat Yeşilyurt

*Department of Software Engineering, Faculty of Computer and Information Sciences,
Sakarya University, 54050 Sakarya, Türkiye
berhat.yesilyurt@ogr.sakarya.edu.tr*

Received (Day Month Year)

Revised (Day Month Year)

Accepted (Day Month Year)

Recognizing isolated handwritten words is difficult because handwriting varies strongly between writers. We study this problem on the IAM database under the Aachen writer-disjoint protocol, reporting every result on all 20,310 test words. Our system, CRNN-LX, is a convolutional-recurrent network trained from scratch and followed by a lexicon-aware post-corrector. Two questions are examined under identical conditions. The first concerns the post-corrector, usually reported as a single dictionary or language-model step. Separating its two ingredients shows that they do different jobs: lexicon coverage decides the sign of the correction (the 7 K-word training vocabulary lowers accuracy by 2.4 points, a 239 K-type general English list raises it by 0.7), while the unigram frequency prior that ranks candidates within edit distance supplies most of the magnitude, a further 1.2 points, and is worth twice as much on the large lexicon. Together they move accuracy from 78.83% to 80.74%. The second question is whether writing-aware augmentation (elastic, morphological and wide photometric transforms) helps. Under a single-seed setting, six configurations differing only in augmentation show no statistically detectable gain from these transforms, whereas removing the conventional pipeline beneath them costs 2.5 points, consistent with saturation rather than an insensitive experiment.

Keywords: Handwritten text recognition; lexicon-based post-correction; unigram fre-

*Corresponding author.

quency prior; convolutional-recurrent neural network; connectionist temporal classification; data augmentation; ablation study.

1. Introduction

Offline *handwritten text recognition* (HTR) is the task of converting a scanned image of handwriting into machine-readable text without access to the pen trajectory. Depending on how the input is segmented, an HTR system may operate on full pages, on text lines, or — as in this work — on isolated *word* images. Word-level recognition is the harder setting per sample, because the recognizer cannot exploit the surrounding words of a sentence to disambiguate an unclear glyph.

The IAM Handwriting Database¹¹ is the standard benchmark for English HTR, containing roughly 115 000 word images from 657 writers. We adopt the partition released by RWTH Aachen University¹⁶ — referred to as the *Aachen split* throughout this paper — which guarantees that no writer contributes to more than one of the training, validation and test sets. This *writer-disjoint* protocol is stricter than a random split and is the realistic condition for deploying a recognizer on handwriting it has never seen.

Modern recognizers are trained with the *connectionist temporal classification* (CTC) objective,⁵ which removes the need for character-level alignment between the image and its transcription. Two model families dominate: convolutional-recurrent networks (CRNNs) trained with CTC,^{1,14,18} and attention-based sequence-to-sequence or transformer encoders.^{8,12} For a compact CRNN, two ingredients are commonly credited with closing part of the gap to heavier models: data augmentation that reflects how handwriting actually varies, and a lexical prior applied at or after decoding. Both are widely used, but their individual contribution is rarely measured under controlled conditions — same data, same architecture, same training schedule, same test set.

This paper provides such a measurement, and finds the two ingredients unequal. On the augmentation side, the object of study is three writing-aware transforms added on top of a conventional geometric and photometric pipeline (rotation, translation, scaling, shear, brightness and contrast jitter, Gaussian noise, gamma correction and random erasing; Section 3.4): *elastic deformation*, which imitates the small variations of a stroke from one writing to the next; *morphological perturbation*, which thins or thickens the strokes as a fine or a broad pen would; and a *widened photometric range*, which imitates lighter or darker ink and scans. Throughout the paper we call these three the *proposed transforms* and everything else the *conventional pipeline*. The label is ours rather than a standard: there is no agreed set of routine augmentations for handwriting, and some prior systems already treat elastic distortion as one.⁴ On the lexical side, the post-corrector replaces a recognized word by a lexicon entry within a small edit distance and ranks the candidates by word frequency. We name the proposed system CRNN-LX (CRNN with lexicon-extended, frequency-ranked post-correction) and its lexicon-free counterpart CRNN-G; Ta-

ble 1 defines the six configurations compared. Our contributions are:

- **An anatomy of lexicon-assisted correction.** Holding the optical model fixed, we vary the post-correction lexicon and the ranking rule independently, which separates two effects that are usually reported as one. Coverage decides the sign: the training vocabulary alone *hurts*, by 2.4 to 2.7 percentage points (pp) on the five augmented models, while a general English list helps (+0.3 to +0.7 pp). The frequency prior then supplies most of the magnitude (+1.1 to +1.3 pp), and its value grows with the lexicon, because a larger candidate set makes the ranking matter more. Neither ingredient alone accounts for the +1.5 to +1.9 pp that the full corrector delivers.
- **A controlled augmentation ablation with a positive control.** Six networks that differ only in the augmentation schedule are trained from scratch on the full Aachen training set and evaluated on the full test set. Elastic deformation, morphological perturbation and a widened photometric range change word accuracy by at most 0.38 pp, and none of those differences is statistically significant; removing the conventional pipeline underneath them costs 2.49 pp at $p = 2 \times 10^{-27}$. These results are consistent with a saturation effect rather than with an inability of the experimental setup to detect augmentation-related improvements.
- **A methodological observation on elastic deformation.** The common parameterization — normalized Gaussian-smoothed noise scaled by $\alpha \in [2, 5]$ — yields root-mean-square (RMS) displacements of only 0.02–0.04 pixels per axis at the working resolution and is effectively a no-op; we report the corrected parameterization we used and its effect.
- **A verified evaluation protocol and full release.** Writer-, prompt- and form-level disjointness of the split are checked programmatically; all 20,310 test words are used; code, split lists and per-word predictions are public.

Section 2 groups the related literature into four groups and positions our work within them. Section 3 describes the pipeline of Fig. 1. Section 4 formalizes the evaluation metrics and statistical tests. Section 5 reports the results, Section 6 analyzes them, and Section 7 concludes.

2. Related Work

We organize prior work into four groups according to how each relates to ours: (i) *recurrent CTC recognizers*, which share our optical backbone but are almost always evaluated on text lines; (ii) *attention and transformer recognizers*, which target the same word-level task as ours with a heavier decoding paradigm; (iii) *other word-level recognizers on IAM*, whose evaluation protocols differ from ours; and (iv) *lexical decoders*, which use a lexicon or language model inside the decoding loop, whereas our post-corrector applies it afterwards. For each group we state what we adopt and where we differ.

Recurrent CTC recognizers. Graves and Schmidhuber⁶ introduced multi-dimensional long short-term memory (LSTM) networks with CTC as an early end-to-end offline recognizer. Puigcerver¹⁴ showed that a one-dimensional bidirectional LSTM (BiLSTM) stacked on a convolutional neural network (CNN) feature extractor matches multi-dimensional recurrence at a fraction of the cost, defining the canonical CRNN baseline that our architecture follows. Bluche and Messina¹ added gated convolutions, and Flor et al.³ combined a CNN with a bidirectional gated recurrent unit (BGRU) and a language-model pipeline. Wigington et al.²¹ studied augmentation for CNN-LSTM recognizers and reported gains from combined transforms on IAM lines. We adopt the CRNN backbone unchanged; our contribution lies in measuring, rather than assuming, the effect of the augmentation and of the lexical prior.

Attention and transformer recognizers. Sueiras et al.²⁰ and Kang et al.⁸ introduced sequence-to-sequence recognizers with content-based attention for isolated IAM words. Kang et al.⁷ later integrated a language model into the decoder itself through candidate fusion, and Kass and Vats⁹ combined a residual-network (ResNet) encoder, attention decoding and transfer learning from scene-text corpora, reporting the strongest word-level accuracy on the Aachen split that we are aware of. These four systems share our task, metric and split and are listed in Section 5. Transformer architectures such as TrOCR¹⁰ have since lowered line-level error rates further, but require pre-training on hundreds of millions of synthetic lines. Multimodal large language models have also been benchmarked on IAM;² those systems operate on full pages and rely on proprietary pre-training corpora, so they are not comparable to a from-scratch word recognizer.

Other word-level recognizers on IAM. Dutta et al.⁴ improved a CNN–recurrent neural network (RNN) hybrid with synthetic pre-training, slant normalization and test-time augmentation and report a 12.61% unconstrained word error rate on IAM words evaluated without punctuation or letter case, a setting not on the same footing as a case-sensitive recognizer trained from scratch. Rajesh et al.¹⁵ re-trained that CNN–RNN together with their own HWRCNet, a CNN–BiLSTM operating in the JPEG discrete cosine transform (DCT) domain, on a random 95:5 partition of IAM and obtained 77.14% and 80.08% strict word accuracy on uncompressed images; we note that their headline figure of 89.05% is *word accuracy with flexibility* of two edit operations rather than strict accuracy. Taking a different route altogether, Mondal et al.¹³ recognize IAM words lexicon-free by detecting individual characters with a YOLOv3 object detector, trained on only 1 200 word images, and report a 29.21% word error rate on 30 000 randomly selected test words. All three systems use partitions of IAM other than ours, a point we return to below.

Lexical decoders. Most systems that exploit a vocabulary do so while decoding. Word beam search (WBS)¹⁷ injects a dictionary constraint directly into CTC

beam search, so that only lexicon words can be produced; candidate fusion⁷ feeds a language model’s proposals into an attention decoder; and Flor et al.³ couple the optical model to a language-model pipeline. In all of these the lexical knowledge is part of decoding and cannot be varied without re-decoding or retraining. We instead apply it *after* greedy decoding, which lets us swap the lexicon and the ranking rule independently while the optical hypotheses stay fixed. That is the property on which the decomposition of Section 5.2 rests. **Positioning.** Of the word-level systems above, only the attention-based recognizers of Sueiras et al.,²⁰ Kang et al.,^{7,8} and Kass and Vats⁹ are evaluated on the Aachen writer-disjoint partition that we use; the CRNN-family results of Rajesh et al.¹⁵ and the detector of Mondal et al.¹³ use their own random partitions of IAM. Comparisons across protocols are indicative only; the controlled comparisons of this paper are those between our own configurations.

3. Proposed System

Fig. 1 shows the complete processing chain, from the raw IAM word crops to the reported metrics, and marks the two stages whose effect this paper measures.

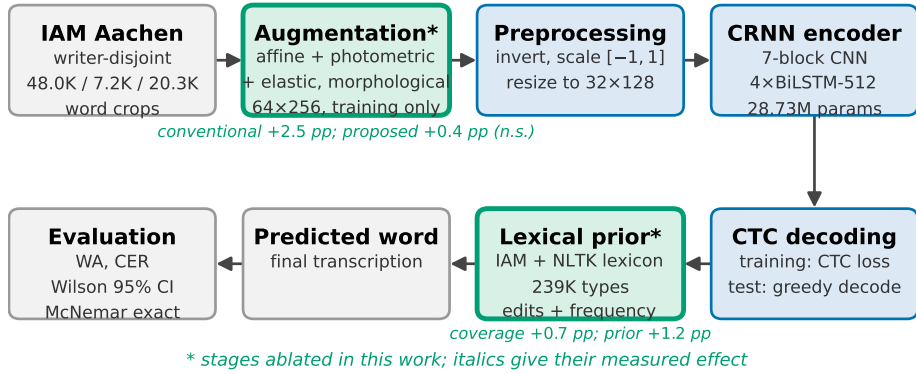


Fig. 1. Overview of CRNN-LX. Word crops are augmented (training only), normalized, encoded by a CRNN and decoded with CTC; the decoded string is then corrected against a lexicon-extended unigram frequency prior. The two outlined stages are the ones varied in the ablations, and each carries the effect measured for it. The evaluation stage reports word accuracy (WA), character error rate (CER) and Wilson confidence intervals (CI), all defined formally in Section 4.

3.1. Data and Protocol

We use the word-level IAM partition of the Aachen split:¹⁶ 747 training forms, 116 validation forms and 336 test forms with disjoint writer sets (283, 55 and 161 writers respectively). Following common practice, only word crops whose segmentation is flagged `ok` are retained. Five validation forms whose prompt text also occurs in the test set were removed from the validation partition so that checkpoint selection is independent of the test prompts; the test partition is used exactly as published. This yields 47 997 training words (two IAM image files are empty and are skipped), 7 205 validation words from 111 forms, and 20,310 test words from all 336 test forms. Writer, prompt and form disjointness between the three partitions, as well as the integrity of every record and image, are verified by a script distributed with the code. The label alphabet contains 78 symbols (upper- and lower-case Latin letters, digits and punctuation), plus the CTC blank.

3.2. Model Naming

Table 1 lists every configuration evaluated. All six share the optical model of Section 3.3, the training schedule and the post-corrector of Section 3.5; they differ only in the augmentation applied during training. CRNN-G is the fully augmented optical model decoded greedily, without any lexicon; CRNN-LX is that same model followed by the extended-lexicon, frequency-ranked post-corrector of Section 3.5, and is the configuration we propose. The suffixes denote the post-processing rather than the augmentation, which turns out not to matter (Section 5.1). The first row is the zero-augmentation anchor: it has the conventional pipeline switched off as well, and exists so that the ablation can distinguish “the proposed transforms add nothing” from “this pipeline cannot detect augmentation at all”.

Table 1. Configurations compared. All use the same 28.73 M-parameter network, the same schedule and the same post-corrector. The rows are cumulative: elastic and morphological perturbation are each added on top of the widened photometric range. Conventional: the pipeline listed in Section 3.4.

Name	Conventional	Wide photometric	Elastic	Morphological
no augmentation	no	no	no	no
CRNN-B (baseline)	yes	no	no	no
+ wide photometric	yes	yes	no	no
+ elastic	yes	yes	yes	no
+ morphological	yes	yes	no	yes
CRNN-LX (proposed)	yes	yes	yes	yes

3.3. Optical Model

The encoder is a seven-block convolutional feature extractor that collapses the input to a single row of 512-dimensional frames, followed by a four-layer bidirectional

LSTM with 512 hidden units and a linear projection onto the 79 output classes; the network has 28.73 M parameters. Inputs are grayscale crops resized to 32×128 , inverted so that ink is high-valued, and scaled to $[-1, 1]$. The network is trained with the CTC objective⁵ using AdamW with learning rate $\eta = 7 \times 10^{-4}$ and weight decay 10^{-5} , five warm-up epochs followed by cosine decay, mixed precision, and batches of 128, for at most 100 epochs. Training stops when neither the validation loss nor the validation word accuracy has improved for 15 epochs, and the checkpoint with the best validation word accuracy is evaluated once on the test set.

3.4. Augmentation

The baseline (CRNN-B) uses what we call the conventional pipeline: rotation ($\pm 7^\circ$), translation ($\pm 5\%$), scaling (0.9–1.1), shear ($\pm 5^\circ$), brightness and contrast jitter (0.85–1.15), Gaussian noise ($\sigma = 0.03$), gamma correction (0.80–1.20) and random erasing. The proposed transforms, which the ablation variants add cumulatively, are:

- **Wide photometric range.** Brightness and contrast 0.70–1.35, gamma 0.70–1.30, noise $\sigma = 0.05$.
- **Elastic deformation** (probability 0.5). A random displacement field is smoothed with a Gaussian kernel whose width is 8% of the larger image side, rescaled to unit RMS, and multiplied by an amplitude drawn uniformly from $[1, 3]$ pixels; the crop is warped accordingly. The unit-RMS normalization is essential: the frequently used variant that multiplies the *unnormalized* blurred noise by $\alpha \in [2, 5]$ ¹⁹ produces, at the 64×256 working resolution, per-axis RMS displacements of only 0.02–0.04 pixels (never more than 0.19 px), i.e. it leaves the image visually and numerically unchanged. Our own earlier implementation had this property.
- **Morphological perturbation** (probability 0.15). The crop is eroded or dilated, with equal probability, by a 2×2 kernel, reproducing the difference between a fine and a broad nib.

Fig. 2 illustrates the transforms. All augmentation is applied on the graphics processing unit (GPU) to whole batches at a working resolution of 64×256 pixels (the median IAM word height is 68 px), with independent random parameters per sample, before the final resize to 32×128 ; the same code path is used by all six configurations. The zero-augmentation anchor bypasses that code path entirely, so its crops reach the network with preprocessing only.

3.5. Lexicon-Extended Post-Correction

After greedy CTC decoding, each hypothesis is checked against a closed lexicon. If the hypothesis (case-insensitively) is in the lexicon it is accepted unchanged; otherwise it is replaced by the best-scoring lexicon entry within a Levenshtein distance

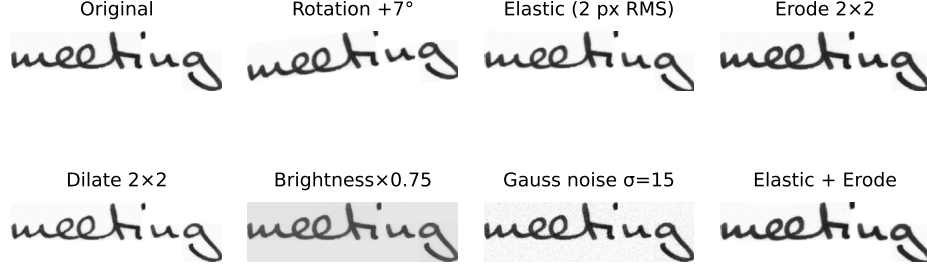


Fig. 2. Augmentation transforms on a training crop. The elastic panel uses the corrected amplitude (2 px RMS); the last panel shows the compound elastic+morphological effect.

of 1 (words of up to four characters) or 2 (longer words). The score is a *unigram frequency prior*: the add-one-smoothed log-probability of the candidate under the word counts of the 47 997 IAM training words (7 173 types), minus a penalty of 5 per edit. The released implementation also stores bigram and trigram counts for use on text lines, but an isolated word image has no preceding words, so only the unigram statistics enter the score; we therefore call the component a frequency prior and avoid the term *n-gram*. Context would in any case be sparse: the two words preceding a test token form a trigram context that occurs in the training text for only 34% of the test words.

The lexicon is the variable of interest. The *training lexicon* consists of the 7 173 word types of the training transcriptions; the *extended lexicon* adds the Natural Language Toolkit (NLTK) English word list, giving 239 126 types. Measured on the test transcriptions, the training lexicon covers 84.8% of the test tokens and the extended lexicon 94.0%; the remaining tokens are proper nouns, inflected forms absent from both lists, and punctuation-bearing strings. Only the word *list* is extended: the frequency counts are estimated on IAM training text alone, so no external corpus statistics enter the model. Section 5 measures the effect of each choice with the optical model held fixed.

4. Experimental Setup

4.1. Evaluation Metrics

Let $\mathcal{T} = \{(\hat{y}_i, y_i)\}_{i=1}^N$ be the set of $N = 20,310$ predicted/reference word pairs. *Word accuracy* (WA) is the exact-match rate

$$\text{WA} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\hat{y}_i = y_i], \quad (1)$$

and the word error rate (WER) follows as $\text{WER} = 1 - \text{WA}$. The *character error rate* (CER) aggregates Levenshtein distances $d(\cdot, \cdot)$ over the corpus,

$$\text{CER} = \frac{\sum_{i=1}^N d(\hat{y}_i, y_i)}{\sum_{i=1}^N |y_i|}, \quad (2)$$

so that a single badly recognized long word cannot dominate the score.

4.2. Confidence Intervals

Because WA is a proportion estimated on a finite test set, we report Wilson 95% confidence intervals (CIs) rather than normal-approximation intervals, which are unreliable near the boundaries. For k correct predictions out of N and $z = 1.96$,

$$\text{CI}_{95} = \frac{\hat{p} + \frac{z^2}{2N} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z^2}{4N^2}}}{1 + \frac{z^2}{N}}, \quad \hat{p} = \frac{k}{N}. \quad (3)$$

With $N = 20,310$ the half-width of the interval is about 0.55 pp.

4.3. Significance Testing

Two systems evaluated on the same test set produce paired outcomes, so we use the exact McNemar test. Let b be the number of samples that the first system recognizes correctly and the second does not, and c the converse. Under the null hypothesis the discordant pairs are distributed $\text{Bin}(b + c, \frac{1}{2})$, giving the two-sided p -value

$$p = \min \left(1, 2 \sum_{i=0}^{\min(b,c)} \binom{b+c}{i} 2^{-(b+c)} \right). \quad (4)$$

We treat $p < 0.01$ as significant.

4.4. Reproducibility

All six configurations were trained on the same machine (one NVIDIA GeForce RTX 4070 Laptop GPU with 8 GB, PyTorch 2.7.1 with CUDA 11.8, Python 3.13) from the same code revision, with seed 42, at 35–75 s per epoch; a configuration takes roughly one hour. Because augmentation is random, a single seed per configuration is used and differences smaller than the confidence interval are not interpreted. Every number reported below comes from one evaluation pass in single precision, in which the six optical models and the five post-corrections are scored against identical inputs; two independent runs of that pass agree on all 20,310 hypotheses. Mixed-precision inference, by contrast, is not bit-reproducible and shifted 2 words between runs (0.01 pp). Code, split definitions, the split-verification script, per-word predictions and evaluation scripts are available at <https://github.com/Ridvan013/CRNN-Handwriting-Recognition> (branch `feature/aachen-v3-extended-trigram`); the trained weights exceed the repository file-size limit and are distributed separately.

5. Results

5.1. Augmentation Ablation

Table 2 is the central result, and it separates into two layers.

The conventional pipeline matters. Removing it costs 2.49 pp (80.36 to 77.87%) and 1.66 pp of character error rate; 1 346 words are recognized only by the baseline against 840 only by the anchor, and the exact McNemar test rejects equality at $p = 2 \times 10^{-27}$. Without a lexicon the same comparison is wider still, 78.76 against 74.33% (−4.43 pp), so part of the weaker optical model is masked by the post-corrector.

Under the reported single-seed setting, the proposed transforms do not provide a statistically detectable improvement. The five configurations that keep the conventional pipeline reach 80.33–80.74% and are statistically indistinguishable: the full scheme (CRNN-LX) is nominally the best, 0.38 pp above the baseline, but that is within one confidence-interval half-width and no pair is rejected ($p \geq 0.060$). Because the configurations are nested, Table 2 reports two differences: Δ_B against the baseline CRNN-B, and Δ_P against the +wide photometric configuration on which elastic and morphological perturbation are built, which isolates each transform. By Δ_P , elastic perturbation adds 0.34 pp, morphological perturbation 0.31 pp and the two together 0.41 pp; the widened photometric range itself changes nothing (−0.03 pp), which is why Δ_B is smaller by at most 0.03 pp (+0.32, +0.28 and +0.38 pp). On the per-word level these five disagree on 3 328 words (16.4%), yet the disagreements cancel: for the baseline–CRNN-LX pair, 777 words are recognized only by the baseline and 854 only by CRNN-LX. Among configurations that already augment, the choice of transform changes *which* words are recognized far more than *how many*.

Table 2. Augmentation ablation on the Aachen writer-disjoint test set ($N = 20,310$). All rows share the network, schedule and post-corrector. Δ_B is the difference in WA to CRNN-B and Δ_P the difference to the +wide photometric configuration on which elastic and morphological perturbation are added; both are computed from unrounded accuracies. p is the exact McNemar test against CRNN-B on per-word outcomes. The first row is the zero-augmentation anchor and acts as a positive control. Best validation WA and the epoch at which it was reached are given for reference (validation: 7 205 words, 55 writers).

Configuration	WA (%)	CER (%)	Wilson 95% CI	Δ_B	Δ_P	p	best val WA (ep.)
no augmentation	77.87	11.00	[77.29, 78.43]	−2.49	—	2×10^{-27}	82.57 (38)
CRNN-B (baseline)	80.36	9.34	[79.81, 80.90]	(ref.)	—	(ref.)	84.61 (87)
+ wide photometric	80.33	9.34	[79.78, 80.87]	−0.03	(ref.)	0.901	84.34 (76)
+ elastic	80.67	9.09	[80.13, 81.21]	+0.32	+0.34	0.122	84.69 (77)
+ morphological	80.64	9.09	[80.09, 81.17]	+0.28	+0.31	0.165	84.58 (84)
CRNN-LX (all)	80.74	9.09	[80.19, 81.28]	+0.38	+0.41	0.060	84.77 (70)

Fig. 3 shows the mechanism. The anchor separates from the rest early: it drives the training loss down to 0.018 — a fifth of what the augmented runs reach — while

its validation loss rises, peaks at 82.57% validation accuracy by epoch 38 and triggers early stopping at epoch 53. It memorises the training set. The five augmented runs keep improving to epoch 70–87 and converge to a plateau of 84.3–84.8%, and among them the curves are indistinguishable after the first fifteen epochs; their training losses differ slightly (the fully augmented model fits the training set a little less tightly, as a regularizer should), but that difference never reaches the validation set. With 47 997 training words from 283 writers, the conventional pipeline already covers the variability that the proposed transforms simulate.

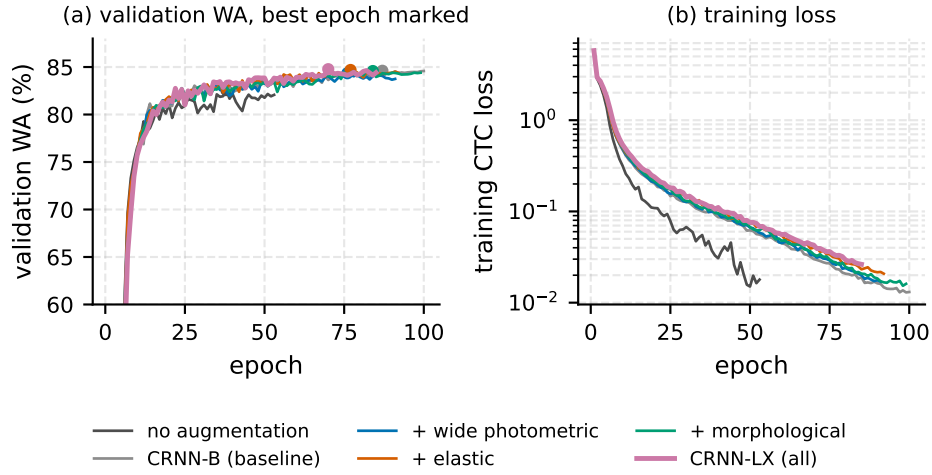


Fig. 3. Training dynamics of the six configurations: (a) validation word accuracy with the best epoch marked, (b) training CTC loss. The zero-augmentation anchor reaches the lowest training loss and the lowest validation accuracy; the five augmented curves coincide.

5.2. Lexicon Ablation

Table 3 holds the optical model fixed (CRNN-LX, best checkpoint) and varies only what happens after greedy decoding. The effect is an order of magnitude larger than that of the augmentation. The corrector has two ingredients — the lexicon a hypothesis is tested against, and the frequency prior used to rank the candidates within edit distance — and the table varies them independently, so that rows 2 and 4 isolate coverage at fixed ranking while rows 2→3 and 4→5 isolate the prior at a fixed lexicon.

Coverage decides the sign. Against the training vocabulary alone, accuracy *falls* from 78.83% to 76.42%: hypotheses that are valid English words absent from the 7K training list — 15% of the test tokens — are forced onto the nearest training word, and most of these substitutions are wrong. Replacing that list with 239 K

types, which covers 94% of the test tokens, turns the same -2.41 pp into $+0.68$ pp (76.42 to 79.51%, a swing of 3.09 pp at identical ranking).

The frequency prior supplies most of the magnitude, and its value grows with the lexicon. On the 7K list it is worth $+0.62$ pp (76.42 to 77.04%); on the 239K list, $+1.23$ pp (79.51 to 80.74%). Fig. 4 shows both effects at once. The reason is combinatorial rather than linguistic: a larger lexicon puts more entries within the same edit distance of a hypothesis, so the choice among equidistant candidates — which is exactly what the prior makes — decides more cases, and Section 6.2 takes the argument further. Neither ingredient alone reaches the $+1.91$ pp of the full corrector. The pattern holds on the other four augmented optical models of Table 2: the training lexicon costs 2.4–2.7 pp, the extended lexicon gains 0.3–0.7 pp over lexicon-free decoding, the prior adds a further 1.1–1.3 pp, and the complete corrector gains 1.5–1.9 pp. On the zero-augmentation anchor, whose greedy output is 4.4 pp weaker, the ordering is unchanged but every effect is larger and the complete corrector gains 3.54 pp: the more the optical model errs, the more a lexical prior has to repair.

Table 3. Post-correction ablation with the optical model fixed (CRNN-LX checkpoint, $N = 20,310$). Each row applies a different post-correction to the *same* greedy hypotheses; lexicon size and frequency ranking vary independently.

Post-correction	Lexicon size	WA (%)	CER (%)	Wilson 95% CI
none (CRNN-G, greedy CTC)	n/a	78.83	8.60	[78.27, 79.39]
training lexicon, edit distance only	7 173	76.42	11.19	[75.83, 76.99]
training lexicon + frequency ranking	7 173	77.04	10.94	[76.46, 77.61]
extended lexicon, edit distance only	239 126	79.51	9.71	[78.95, 80.06]
extended lexicon + frequency ranking (CRNN-LX)	239 126	80.74	9.09	[80.19, 81.28]

The character error rate moves the other way: the lexicon-free output has the lowest CER (8.60%), and every corrector raises it. A whole-word corrector can only replace a hypothesis by a complete lexicon entry; when the replacement is wrong, a near-miss with one character error becomes a different word with several. The training lexicon, which replaces many correct-but-unlisted words, pays this price most heavily (11.19%); the extended lexicon pays it rarely (9.09%) but still raises CER by half a point while improving WA by almost two. Lexicon-assisted word accuracy and character error rate therefore measure different things and should be reported together.

5.3. Comparison with Prior Word-Level Systems

Table 4 places CRNN-LX next to seven published word-level systems and states, for each, whether a lexicon or language model was applied at test time. The upper block lists results obtained on the Aachen writer-disjoint partition. The closest match in protocol is the sequence-to-sequence recognizer of Sueiras et al.:²⁰ the sizes of their

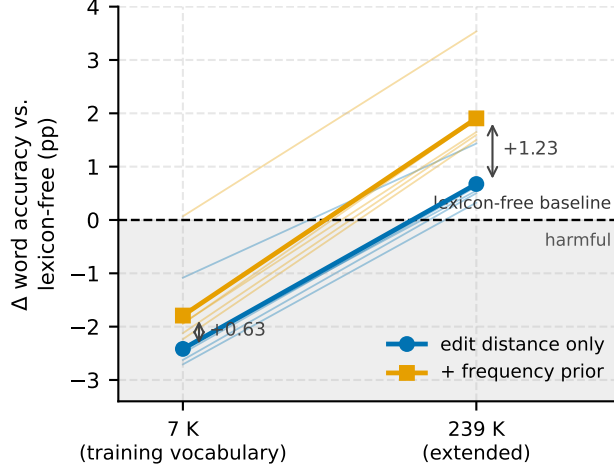


Fig. 4. What the post-corrector contributes, measured against lexicon-free decoding. Coverage decides the sign: with 7 K types both variants fall below the baseline, with 239 K both rise above it. The vertical gap between the two lines is the contribution of the frequency prior, and it doubles as the lexicon grows. Thick lines: the CRNN-LX optical model; faint lines: the other five configurations of Table 2.

partition (47 952, 20 306 and 7 558 words, as compiled in¹³) are those of the ok-filtered Aachen training, test and validation lists (47 999, 20 310 and 7 559) from which ours are drawn, and they report lexicon-free word and character error rates on the full test partition exactly as we do. Under lexicon-free decoding CRNN-G is 2.6 pp more accurate (78.83% against 76.20%) at a slightly lower CER (8.60% against 8.80%); with the extended lexicon CRNN-LX is 4.5 pp ahead. CRNN-LX remains below the later attention-based systems: 1.8 pp below Kang et al.,⁸ 3.4 pp below the candidate-fusion model of Kang et al.⁷ and 3.9 pp below AttentionHTR,⁹ which additionally transfers knowledge from 14.4 M scene-text images; against our lexicon-free figure each gap is 1.9 pp wider. The gap is larger at character level (5.79–6.88% against our 8.60–9.09%), consistent with the autoregressive decoders of those systems modeling inter-character dependencies that a CTC output with whole-word post-correction cannot. The lower block reports three systems measured on other partitions of IAM; CRNN-LX is above all three and CRNN-G above two, but the partitions differ in writer overlap and size and we draw no conclusion from margins of this size.

5.4. Error Analysis

CRNN-LX misrecognizes 3 912 of the 20,310 test words. Only 274 of them (7.0%) differ from the reference in letter case alone; the remaining errors are genuine. At character level the aligned errors consist of 5 244 substitutions, 1 804 deletions and

Table 4. Word-level results on IAM. WA in parentheses is derived as $1 - \text{WER}$ from the cited paper. Lexicon: lexical constraint at test time (none = unconstrained decoding; LM = language model without a closed lexicon; n/s = not stated).

System	Year	Lexicon	WA (%)	CER (%)
<i>Aachen writer-disjoint partition (≈ 20.3 K test words)</i>				
Sueiras et al. ²⁰	2018	none	(76.20)	8.80
Kang et al. ⁸	2018	none	(82.55)	6.88
Kang et al. ⁷	2021	LM	(84.09)	5.79
Kass and Vats ⁹	2022	none	(84.60)	6.50
CRNN-G (ours)	2026	none	78.83	8.60
CRNN-LX (ours)	2026	239 K general	80.74	9.09
<i>Other partitions of IAM</i>				
Mondal et al. ^{13a}	2022	none	(70.79)	9.53
CNN-RNN of, ⁴ re-trained in ^{15b}	2022	n/s	(77.14)	11.08
Rajesh et al. ^{15b}	2022	n/s	80.08	9.89

^a30 K randomly selected test words; 1 200 training images.

^bRandom 95:5 split, writer-dependent, ≈ 5.8 K test words.

394 insertions, so the model drops characters far more often than it hallucinates them. Fig. 5 shows the ten most frequent substitutions: they are symmetric confusions between visually similar lower-case glyphs — $\text{o} \leftrightarrow \text{a}$ (242 cases), $\text{s} \leftrightarrow \text{r}$ (150), $\text{n} \leftrightarrow \text{r}$ (98), $\text{n} \leftrightarrow \text{m}$ (98), $\text{t} \leftrightarrow \text{l}$ (97), $\text{e} \leftrightarrow \text{a}$ (86). Case substitutions account for 7.7% of all substitutions. These are errors of shape, not of vocabulary: a whole-word lexicon can repair them only when the mis-shaped word falls outside the lexicon and its correct form lies within two edits, which is exactly the population that the extended lexicon recovers in Table 3.

6. Discussion

6.1. Interpretation of the Augmentation Results

The anchor rules out the explanation that would otherwise be hardest to exclude. A null result can always mean that the experiment is too noisy to see anything, and with a single seed per configuration that objection has force. But the same test, the same data and the same seed detect the removal of the conventional pipeline at $p = 2 \times 10^{-27}$, two and a half points of word accuracy. The apparatus is not blind to augmentation; it is blind to *these* transforms on *this* training set. What we observe is consistent with saturation, and three further observations indicate where it occurs.

First, the anchor overfits and the augmented runs do not: training loss 0.018 against 0.09–0.11, early stopping 30 epochs sooner (Fig. 3). The first increment of augmentation buys regularization that the network genuinely needs. Second, among the augmented configurations the validation curves coincide and the extra transforms neither slow convergence nor raise the plateau, while the fully augmented model reaches a slightly higher training loss — so they do regularize further, but

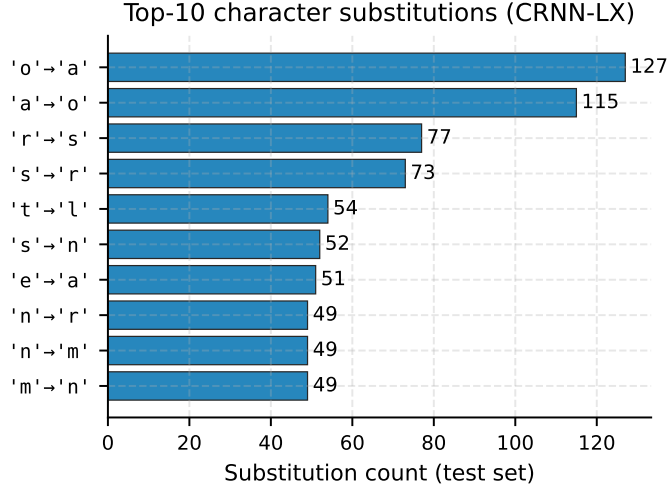


Fig. 5. Ten most frequent character substitutions of CRNN-LX on the test set (true → predicted).

the baseline is no longer overfitting in a way that costs accuracy on unseen writers. Third, the per-word disagreements between those configurations are large yet balanced. Together these indicate that with 48 K training words from 283 writers, the writer-to-writer variation already present in the data exceeds what elastic and morphological perturbation of 64-pixel-high crops can add on top of affine and photometric jitter. Reports of sizeable gains from such augmentation, including our own preliminary experiments, came from smaller training sets or from comparisons in which augmentation was not the only difference. The result does not imply that augmentation is useless for HTR in general — Wigington et al.²¹ report gains on lines with a different transform family, and our own anchor shows the first increment is worth 2.5 pp — but it does show that further transforms must be measured under controlled conditions before being claimed.

6.2. Coverage and the Frequency Prior

The ablation separates two effects that are usually conflated under “adding a language model”, and they turn out to do different jobs. Membership testing against a list that covers 85% of the test tokens is harmful, because the corrector cannot distinguish a misrecognized word from a correctly recognized word it has never seen and treats both as errors; raising coverage to 94% is what turns the loss into a gain. That gain is small on its own, however. What converts it into the full +1.9 pp is the prior that ranks the candidates, and it is worth twice as much on the large lexicon as on the small one.

At the word level the prior cannot be doing linguistic work: an isolated crop

offers no preceding words, so only the unigram frequency is available. What it does instead is break ties. The tighter the lexicon, the fewer entries lie within two edits of a hypothesis and the less the ranking matters; at 239 K types a mis-shaped word typically has several equidistant neighbours, and picking the frequent one is right more often than picking an arbitrary one. Edit distance identifies the candidate set and frequency chooses inside it: reporting either without the other describes only half of the corrector.

A practical consequence is that word-level HTR results obtained with a closed lexicon should state both the lexicon’s coverage of the test vocabulary and the rule used to rank candidates, without which the reported accuracy is not interpretable. The remaining 6% of uncovered tokens (proper nouns, rare inflections, punctuation-bearing strings) bound what any whole-word corrector can achieve; addressing them requires either an open-vocabulary character-level model or sentence context.

6.3. Comparison with Previous Word-Level HTR Systems

Among the systems of Table 4, the recognizer of Sueiras et al.²⁰ is the one whose protocol matches ours most closely: isolated IAM words, the Aachen writer-disjoint partition drawn from the same ok-filtered word lists — the only prior system for which we could verify this from the reported partition sizes (Section 5.3); we additionally discard two unreadable images and five validation forms whose prompts recur in the test set — and lexicon-free word and character error rates on the full test partition. Architecturally the two differ in every stage: they extract features from a sliding window of image patches with a LeNet-5-style convolutional network and transcribe them with a recurrent encoder–decoder using attention, whereas CRNN-G is a single convolutional-recurrent network trained with CTC. Under lexicon-free decoding CRNN-G reaches 78.83% against their 76.20% (23.8% word error rate), a margin of 2.6 pp that is almost five times the half-width of our confidence interval, at a slightly lower character error rate (8.60% against 8.80%). Post-correction with the extended lexicon widens the word-level margin to 4.5 pp. Sueiras et al. also report lexicon-constrained variants of their recognizer; those depend on the lexicon employed and are not compared here. We read the comparison as evidence that a plain CRNN trained from scratch on the 48 K Aachen training words is competitive with an early attention model under the same protocol, not as evidence about augmentation: under our single-seed setting, Table 2 shows no measurable contribution of the proposed transforms to the margin.

HWRCNet,¹⁵ the closest system in construction (a CNN, a bidirectional LSTM and a CTC layer), reports 80.08% strict word accuracy, 0.66 pp below CRNN-LX and 1.25 pp above CRNN-G. It is measured on a random 95:5 partition whose test set is writer-dependent and about 3.5 times smaller than ours, so neither difference is interpretable.

6.4. Threats to Validity

Each configuration was trained once. The augmentation differences of 0.0–0.4 pp are below what a single seed can resolve, which is why we report them as non-significant rather than as small gains; establishing an effect of that size would require several seeds per configuration. The 2.49 pp effect of the conventional pipeline is an order of magnitude larger than that resolution limit, so the positive control does not depend on the seed in the way the null result does. The lexicon and ranking effects (−2.4 to −2.7 pp for the training vocabulary, +1.1 to +1.3 pp for the frequency prior) are reproduced in sign and ordering on all six independently trained models and are well outside the interval, although their magnitude grows as the optical model weakens. They are, however, measured with one corrector and one word list; a different edit-distance bound or a different general vocabulary would change their size, though the ordering of the five post-corrections is unlikely to change. Validation accuracy exceeds test accuracy by about four points for every configuration; the validation writers are used for checkpoint selection and are only 55 in number, so this gap is expected and does not indicate a fault. The external systems in Table 4 were trained and evaluated by their authors with their own preprocessing and, in the lower block, on a different partition; we therefore treat only our own six configurations as a controlled comparison. Finally, the extended lexicon is a general English word list and contains no IAM test transcriptions, but it is a closed lexicon nonetheless, and CRNN-LX should be read as a lexicon-assisted result.

7. Conclusion

CRNN-LX recognizes isolated handwritten words with 80.74% accuracy (9.09% character error rate) on the complete Aachen writer-disjoint test set of IAM, with a plain CRNN trained from scratch and no external image data. Under the same partition and lexicon-free decoding it is 2.6 pp more accurate than the sequence-to-sequence recognizer of Sueiras et al.²⁰ and 1.8–3.9 pp below the later attention-based systems listed in Table 4. Under controlled, single-seed conditions, elastic deformation, morphological perturbation and a widened photometric range produce no statistically detectable change in this figure, while removing the conventional augmentation pipeline beneath them costs 2.49 pp — a pattern consistent with the benefit of augmentation saturating before the transforms we proposed; a widely used elastic parameterization, moreover, does not deform the image at all. The post-correction stage, by contrast, moves accuracy by two points in either direction: its lexicon’s coverage of the test vocabulary decides whether the correction helps or hurts, and the frequency prior that ranks the candidates supplies most of the gain once coverage is sufficient. Both improve word accuracy at the expense of character error rate.

For word-level HTR the practical conclusions are that augmentation gains should be demonstrated, not assumed, on the full training set, and that an ablation reporting no effect should include the configuration in which the effect is

known to exist; that lexicon coverage and candidate ranking should be reported separately alongside lexicon-assisted accuracy; and that word accuracy and character error rate should be given together whenever a whole-word corrector is used. Future work will test whether the augmentation becomes useful in the low-data regime, replace the whole-word corrector with a character-level open-vocabulary model to address the uncovered 6% of tokens, and evaluate the pipeline on text lines, where the preceding words would allow a genuine bigram or trigram prior.

Acknowledgments

We thank the maintainers of the IAM Handwriting Database and of the Aachen split.

References

1. T. Bluche and R. Messina, Gated convolutional recurrent neural networks for multilingual handwriting recognition, in *14th IAPR International Conference on Document Analysis and Recognition (ICDAR)*, Vol. 1 (IEEE, 2017), pp. 646–651.
2. G. Crosilla, L. Klic and G. Colavizza, Benchmarking large language models for handwritten text recognition, *Journal of Documentation* (2025).
3. A. F. de Sousa Neto, B. L. D. Bezerra, A. H. Toselli and E. B. Lima, HTR-Flor++: a handwritten text recognition system based on a pipeline of optical and language models, in *Proceedings of the ACM Symposium on Document Engineering (DocEng)* (2020).
4. K. Dutta, P. Krishnan, M. Mathew and C. V. Jawahar, Improving CNN-RNN hybrid networks for handwriting recognition, in *16th International Conference on Frontiers in Handwriting Recognition (ICFHR)* (IEEE, 2018), pp. 80–85.
5. A. Graves, S. Fernández, F. Gomez and J. Schmidhuber, Connectionist temporal classification: labelling unsegmented sequence data with recurrent neural networks, in *Proceedings of the 23rd International Conference on Machine Learning (ICML)* (2006), pp. 369–376.
6. A. Graves and J. Schmidhuber, Offline handwriting recognition with multidimensional recurrent neural networks, in *Advances in Neural Information Processing Systems (NeurIPS)* (2008), pp. 545–552.
7. L. Kang, P. Riba, M. Villegas, A. Fornés and M. Rusiñol, Candidate fusion: integrating language modelling into a sequence-to-sequence handwritten word recognition architecture, *Pattern Recognition* **112** (2021) p. 107790.
8. L. Kang, J. I. Toledo, P. Riba, M. Villegas, A. Fornés and M. Rusiñol, Convolve, attend and spell: an attention-based sequence-to-sequence model for handwritten word recognition, in *German Conference on Pattern Recognition (GCPR)*, LNCSVol. 11269 (Springer, 2018), pp. 459–472.
9. D. Kass and E. Vats, AttentionHTR: handwritten text recognition based on attention encoder-decoder networks, in *15th IAPR International Workshop on Document Analysis Systems (DAS)*, LNCSVol. 13237 (Springer, 2022), pp. 507–522.
10. M. Li, T. Lv, J. Chen, L. Cui, Y. Lu, D. Florencio, C. Zhang, Z. Li and F. Wei, TrOCR: transformer-based optical character recognition with pre-trained models, in *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37(11) (2023), pp. 13094–13102.

11. U.-V. Marti and H. Bunke, The IAM-database: an English sentence database for of-line handwriting recognition, *International Journal on Document Analysis and Recognition* **5**(1) (2002) 39–46.
12. J. Michael, R. Labahn, T. Grüning and J. Zöllner, Evaluating sequence-to-sequence models for handwritten text recognition, in *International Conference on Document Analysis and Recognition (ICDAR)* (IEEE, 2019), pp. 1286–1293.
13. R. Mondal, S. Malakar, E. H. Barney Smith and R. Sarkar, Handwritten English word recognition using a deep learning based object detection architecture, *Multimedia Tools and Applications* **81**(1) (2022) 975–1000.
14. J. Puigcerver, Are multidimensional recurrent layers really necessary for handwritten text recognition?, in *14th IAPR International Conference on Document Analysis and Recognition (ICDAR)*, Vol. 1 (IEEE, 2017), pp. 67–72.
15. B. Rajesh, A. K. Gupta, A. Raj, M. Javed and S. R. Dubey, HWRCNet: Handwritten word recognition in JPEG compressed domain using CNN-BiLSTM network, in *International Conference on Data Analytics and Learning (DAL)* (2022).
16. RWTH Aachen University, IAM handwriting database — Aachen writer-disjoint split Widely used community split, distributed with the RWTH Aachen speech and language processing toolkits, (2010).
17. H. Scheidl, S. Fiel and R. Sablatnig, Word beam search: a connectionist temporal classification decoding algorithm, in *16th International Conference on Frontiers in Handwriting Recognition (ICFHR)* (IEEE, 2018), pp. 253–258.
18. B. Shi, X. Bai and C. Yao, An end-to-end trainable neural network for image-based sequence recognition and its application to scene text recognition, *IEEE Transactions on Pattern Analysis and Machine Intelligence* **39**(11) (2017) 2298–2304.
19. P. Y. Simard, D. Steinkraus and J. C. Platt, Best practices for convolutional neural networks applied to visual document analysis, in *7th International Conference on Document Analysis and Recognition (ICDAR)* (IEEE, 2003), pp. 958–963.
20. J. Sueiras, V. Ruiz, A. Sanchez and J. F. Velez, Offline continuous handwriting recognition using sequence to sequence neural networks, *Neurocomputing* **289** (2018) 119–128.
21. C. Wigington, S. Stewart, B. Davis, B. Barrett, B. Price and S. Cohen, Data augmentation for recognition of handwritten words and lines using a CNN-LSTM network, in *14th IAPR International Conference on Document Analysis and Recognition (ICDAR)* (IEEE, 2017), pp. 639–645.